

Letter of Motivation

Introduction

I am **Swathi, Ramesh** and here I write to demonstrate my strong passion for pursuing a **Master of Science in Computer Science** program at **TU Hamburg**. I have a good academic background and relevant technical experience and am excited to further advance my knowledge and technical skills in the core of computer science through this world-class program.

Academic, Professional Experience, Technical Certifications, and Technical Skills

I hold a Bachelor of Engineering in **Computer Science and Engineering** from Visvesvaraya Technological University, India, with a **CGPA of 7.71 (German Grade: 2.3)**. I have a strong grounding in such courses as C Programming for Problem Solving, Data Structures and Applications, Computer Network and Security, Database Management System and Digital Electronics. Prior to this, I completed my Pre-University (12th Class) with a score of **78% (German Grade: 2.0)** and my 10th class exam mark sheet from the Indian Certificate of Secondary Education with **87.84 %** passing marks. I have gained practical experience through several internships. At **Varcons Technologies Pvt Ltd.**, I worked on **Twitter Sentiment Analysis** during the COVID-19 pandemic, where I applied **machine learning models** to analyze real-time data and presented a comprehensive report on the findings. Additionally, I interned at **Karnataka German Technical Training Institute**, where I completed a **Red Hat Certified Engineer (RHCE)** program, equipping me with valuable skills in Linux systems and network administration. My **Bachelor's project**, titled "**Design and Fabrication of Smart Nursing Robot**," aimed at addressing the growing demand for **smart healthcare systems** by leveraging **AI and machine learning** technologies to assist elderly and disabled individuals with personalized healthcare services. In terms of **certifications**, I have successfully completed the **Java Full Stack course** from **Pentagon Space**, a course on **IoT Edge Computing and IoT Analytics** from **Infosys**, and a training program in **Red Hat (RHCE)** from **Karnataka German Technical Training Institute Bengaluru**. These experiences, combined with my **technical skills** in **Network Analysis and Synthesis, Industrial Automation, Digital Electronics, Telecom**, and familiarity with **IoT and machine learning**, have significantly enhanced my competencies and deepened my motivation to pursue advanced studies in **Artificial Intelligence**. Also, I Completed 16 years of Education in the English Language.

Why Master of Science in Computer Science?

The reasons for choosing the **Master of Science in Computer Science** at **TU Hamburg** are explained by its high emphasis placed on practical, technical, and theoretical aspects of Computer Science. As a program that offers Intelligence Engineering, Computer, and Software engineering besides advanced Mathematics the program gives me specific areas of focus such as machine learning and big data- discretion that forms the core of digitalization. The ability to select my subjects and do project work also allows me to develop interdisciplinary applicability through research, which also improves my problem-solving abilities. This also presents great employment opportunities in information system software, medical-related fields, and port technology upon my completion of this program and in the meantime be well equipped and prepared for a doctorate and leadership roles within the information technology sector. Another objective of this program is the international orientation of the program, with courses taught predominantly in English, which will also help me to engage in a global learning environment, providing me with expanded perspectives on emerging technologies. Finally, practical orientation and a more or less equal theory-practice allocation at **TU Hamburg** will prepare me to achieve this goal and allow me to introduce new developments in computer science to the world.

Why Germany?

Germany is perennially a technological innovation powerhouse as well as an engineering excellence engine. Top-tier education, heavy on research and practical applications, is provided by its universities. With a thriving tech industry and a strong focus on automation and AI, it's an ideal country for practicing firsthand. Also, German universities provide several collaborations with leading companies to solve real-world problems for students. Besides, the environment of Germany will be multicultural, and I will be able to interact with different diverse perspectives, which will help me to enrich my learning. I am also interested in the sustainability and advancements in technology to which **Germany** remains committed, which are the same values I have in the career I've chosen.

Why TU Hamburg

Technische Universität Hamburg is known for its engineering and technology focus, and having a first-class computer science program. I also consider that the focus of the university on research, especially in the fields of artificial intelligence and machine learning, is in harmony with my academic interests and my aspirations in the academic field. The proximity to the industry will bring a lot of opportunities from internships, collaborative projects, and so on to make me gain practical experience while studying at **TU Hamburg**. I'll have the benefit of working on innovative projects, and from the university's modern infrastructure and access to cutting-edge technology, contribute to groundbreaking research. Tu Hamburg will also offer an international environment in which I will get a chance to work with other peers from around the world and this makes my academic experience stimulating and diverse from across the world. I am sure **TU Hamburg** is the right place for me to reach my academic and professional goals.

Conclusion

Therefore, I am assured that studying for the **Master of Science in Computer Science** at **Technische Universität Hamburg** will enable me to attain my career goals and greatly impact the information technology sector. I am looking forward to accepting the responsibilities and prospects that this course will bring and further, I am ready to join the **TU Hamburg** family.

Sincerely,

Swathi, Ramesh